

INTERNSHIP REPORT

Submitted in partial fulfilment of the requirements

for the degree of

BACHELOR OF TECHNOLOGY

COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)

Submitted To

School of Computer Science and Engineering



IILM University, Greater Noida, Uttar Pradesh

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Academic Session: 2026–27

CANDIDATE'S DECLARATION

I, KALASH PRIYA, hereby declare that the internship report titled “Data Science Internship at Unlox” has been prepared by me in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering (Data Science) at IILM University, Greater Noida.

The report presents the learning activities, technical work, projects, and outcomes associated with my internship at Unlox. The technical descriptions and project results included in this report are based on the work and project material available to me. Wherever external material has been consulted, it has been acknowledged appropriately.

I further declare that this report has not been submitted to any other institution for the award of any degree or diploma.

Signature: _____

Student Name: KALASH PRIYA

Roll No.: 25SCS1003005309

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported me during my internship and helped me develop a stronger understanding of data science and machine learning. The internship provided an opportunity to connect academic concepts with practical implementation and to work on a range of programming, data handling, visualization, SQL, dashboarding, and machine learning activities.

I am grateful to the mentors and learning team at Unlox for providing guidance, structured learning, practical assignments, and project-oriented exposure. The progression from programming and data handling fundamentals to machine learning concepts helped me understand how data can be transformed into useful analytical and predictive solutions.

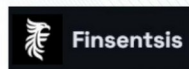
I also thank the faculty members of the School of Computer Science and Engineering, IILM University, for their academic guidance and continuous encouragement. Finally, I express my gratitude to my parents and everyone who supported me throughout the internship period.

Signature: _____

Student Name: KALASH PRIYA

Roll No.: 25SCS1003005309

INTERNSHIP COMPLETION CERTIFICATE



CERTIFICATE

OF INTERNSHIP COMPLETION

THIS IS TO CERTIFY THAT

Kalash Priya

Has successfully completed the Internship & Project-based Learning Program as part of the Job Bridge initiative and has been awarded the Unlox Edge Certification in collaboration With the Finsetis in Data Science on 26-04-2026

UNXDS-INT-3475



JAYANTH MP
VICE PRESIDENT

ABSTRACT

This report presents the learning and practical work completed during a Data Science and Machine Learning internship at Unlox. The internship involved progressive learning from Python programming and data handling fundamentals to data analysis, visualization, SQL, dashboarding, and machine learning. Practical work included a Python calculator, a student database exercise using SQL, a Power BI dashboard, and machine learning projects involving house price prediction and breast cancer diagnosis.

The machine learning work provided hands-on exposure to both regression and classification problems. The House Price Prediction project applied regression analysis to housing attributes and reported an R^2 score of 0.456. The Breast Cancer Classification project used the Breast Cancer Wisconsin (Diagnostic) dataset and reported an accuracy of 98.25%, with strong precision, recall, and F1-scores for both classes. These projects helped reinforce the complete workflow of dataset preparation, exploratory analysis, model training, evaluation, visualization, and interpretation.

Overall, the internship strengthened programming, analytical thinking, machine learning, data visualization, database, and problem-solving skills, while providing practical experience in applying classroom concepts to structured data-driven projects.

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1. INTRODUCTION

1.1 Internship Overview

The internship at Unlox was undertaken with a focus on data science and machine learning. The learning path combined foundational programming with practical data-oriented activities. The work moved from basic Python programming and data manipulation to SQL, visualization, dashboarding, supervised machine learning, model evaluation, and project documentation.

The internship was particularly relevant to the Data Science specialization of the B.Tech CSE program because it provided opportunities to work with datasets, formulate analytical problems, train predictive models, evaluate performance, and communicate results using visualizations.

1.2 Purpose of the Internship

The primary purpose of the internship was to gain practical exposure to data science workflows and machine learning techniques. It also aimed to develop the ability to translate theoretical concepts into working programs and analytical projects.

1.3 Internship Learning Progression

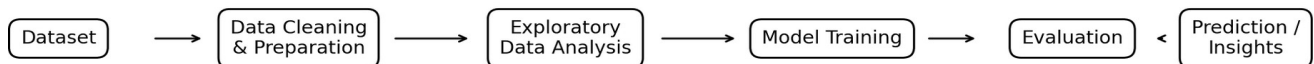


Figure 1.1: General data science and machine learning workflow followed across the practical projects.

The learning progression followed a practical cycle: understanding the dataset or problem, preparing the data, exploring patterns, applying an appropriate analytical or machine learning technique, evaluating the result, and interpreting the output.

2. ORGANIZATION PROFILE

2.1 About Unlox

Unlox operates in the student learning, career development, internship, and industry-readiness ecosystem. Its publicly described programs include domains such as Artificial Intelligence, Data Science, Data Analytics, UI/UX Design, Web Development, Human Resources, and Digital Marketing. The organization also provides project-oriented learning and internship-readiness activities.

The Unlox Data Science learning pathway includes Python, exploratory analysis, NumPy, Pandas, visualization and statistics, supervised learning, unsupervised learning and AI tools, deep learning and NLP, and project-based learning. This structure aligns closely with the practical learning progression followed during the internship.

2.2 Relevance to the Internship

The organization's project-oriented learning approach provided a suitable environment for developing practical data science skills. The internship work combined programming exercises, data management, visualization, and predictive modeling rather than focusing on only one technical area.

2.3 Internship Role

Internship role: Data Science Intern

3. INTERNSHIP OBJECTIVES AND LEARNING PLAN

3.1 Objectives

- Strengthen Python programming and problem-solving skills.
- Understand data collection, cleaning, transformation, and exploratory analysis.
- Develop practical familiarity with NumPy, Pandas, and data visualization.
- Learn SQL concepts including database creation, table design, insertion, selection, filtering, and sorting.
- Develop dashboards and communicate analytical information visually.
- Understand supervised machine learning and apply regression and classification algorithms.
- Learn to evaluate models using appropriate performance metrics.
- Build portfolio-ready projects and document the implementation process.
- Develop the ability to interpret results and identify areas for improvement.

3.2 Learning Areas

Learning Area	Practical Exposure	Outcome
Python Programming	Calculator and programming exercises	Improved control flow, input handling and logic building
SQL	Student database and queries	Understanding of relational database operations
Data Visualization	Graphs and dashboard work	Improved ability to communicate patterns
Machine Learning	Regression and classification projects	Practical model-building and evaluation
Documentation	GitHub project documentation	Improved technical communication

4. TOOLS, TECHNOLOGIES AND LEARNING AREAS

The practical work covered a combination of programming, data analysis, database, visualization, and machine learning tools. The exact software versions used during the internship should be added if required by the university.

- Python – programming, data processing, and machine learning implementation.
- NumPy – numerical operations and array-based computation.
- Pandas – data loading, cleaning, transformation, and analysis.
- Matplotlib / visualization tools – charts and graphical analysis.
- Scikit-learn – machine learning datasets, model training, and evaluation.
- SQL / MySQL-style syntax – relational database creation and querying.
- Power BI – dashboard and business/data visualization.
- Microsoft Excel – spreadsheet-based data handling.
- Git and GitHub – project storage, documentation, and versioned portfolio presentation.
- Jupyter/Google Colab or VS Code – development and execution environment where applicable.

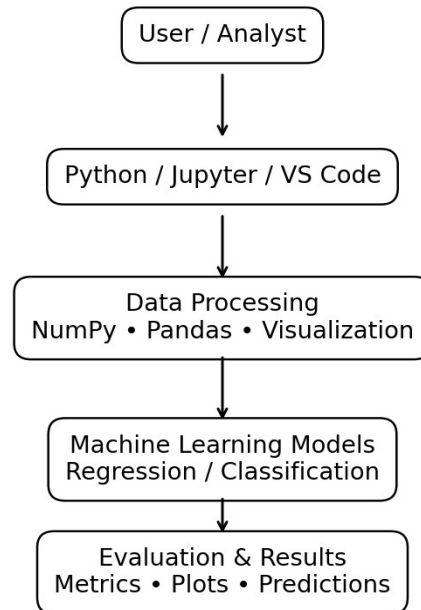


Figure 4.1: Architecture of the practical data science and machine learning workflow.

5. PROJECTS AND WORK PERFORMED

The internship included multiple practical projects and exercises. Together, these activities provided progressive exposure to programming, database management, data visualization, and machine learning.

5.1 Python Calculator

A Python calculator was developed as a programming exercise. The program accepts an operation from the user and supports arithmetic operations including addition, subtraction, multiplication, division, modulus, and exponentiation. It also includes input validation for operations that require a specific number of values and checks for division or modulus by zero.

The project strengthened understanding of conditional statements, loops, lists, user input, numeric data types, arithmetic operators, and basic error-handling logic.

Key Features

- Addition of multiple values
- Subtraction of two values
- Multiplication of multiple values
- Division with zero-denominator validation
- Modulus operation
- Exponentiation
- Basic input validation and user messages

5.2 SQL Student Database

A relational database exercise was completed using SQL. The database was named `student_course_db` and included a `students` table with fields for student identification, name, email, phone number, course, date of birth, and registration date.

The exercise covered database creation, table creation, primary and unique constraints, insertion of records, viewing tables, selecting specific columns, sorting records, ordering registration dates, and filtering students based on course criteria.

Main SQL Operations Practiced

- CREATE DATABASE and USE
- CREATE TABLE with constraints
- INSERT INTO

- SELECT * and selected columns
- ORDER BY ascending and descending
- WHERE filtering
- NOT conditions
- Working with date and datetime fields

The SQL exercise provided practical understanding of how structured information is stored and retrieved from a relational database and created a foundation for data preparation tasks used later in data science.

5.3 EXCEL – BASIC AND ADVANCED LEARNING

During the internship, Excel was studied from foundational to advanced levels, with emphasis on practical spreadsheet-based data handling, calculation, analysis, and presentation. The uploaded workbook demonstrates work with dates and text functions, conditional logic, statistical functions, salary calculations, data classification, concatenation, and basic analytical operations.

Basic Excel Skills

- Entering and organizing tabular data
- Working with rows, columns, cells, and ranges
- Basic arithmetic and formulas
- Sorting and filtering data
- Working with dates and months
- Using COUNT, SUM, AVERAGE, MIN, and MAX
- Formatting and presenting structured data

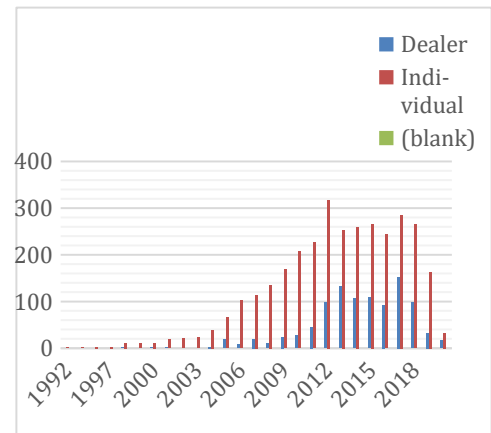


Figure 5.3 Applying formula-based data analysis and interpretation

Advanced Excel Skills

- Using IF statements for conditional classification
- Using SUMIF for conditional aggregation
- Creating calculated salary fields such as incentive, TDS, PF, and net salary
- Combining text fields with CONCATENATE
- Generating formatted email strings from first and last names

5.4 Power BI Dashboard

A Power BI report/dashboard project was also completed. The available project file contains a dataset using booking and revenue-related data, including a visual based on average/summarized revenue over a date hierarchy. This work provided exposure to dashboard construction, data modeling concepts, visual analytics, and presenting trends in an interactive reporting environment.

The dashboard activity complemented the machine learning work by developing the ability to communicate data insights visually before and alongside predictive modeling.

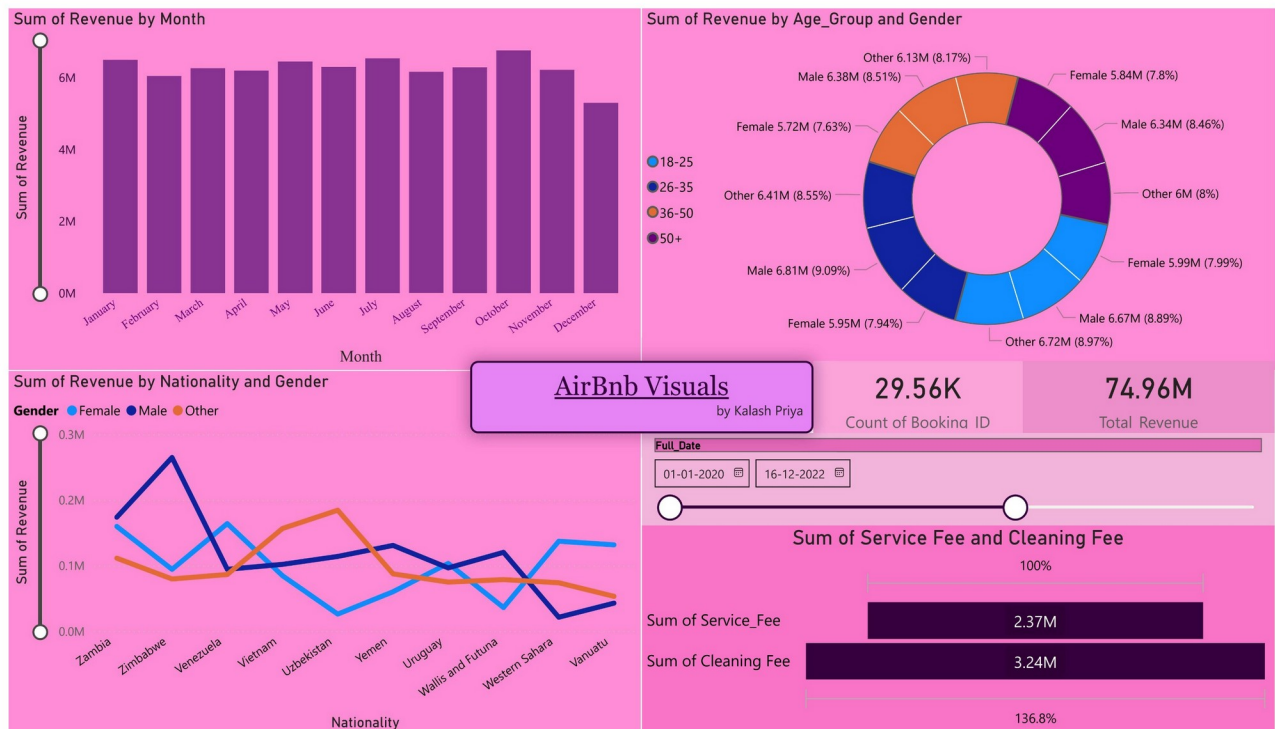
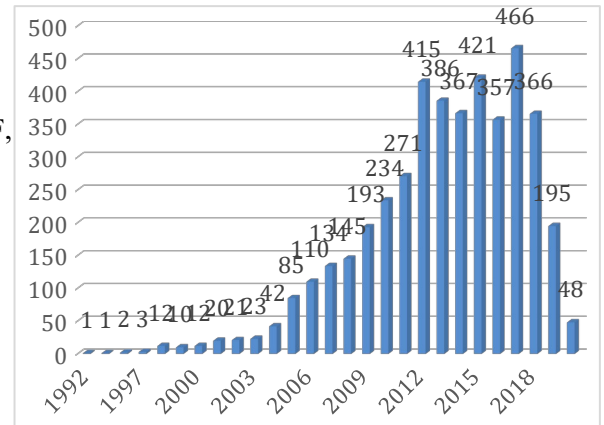


Figure 5.4 The actual AirBnb dashboard is reproduced below from the submitted dashboard PDF so that the report documents the visual output of the internship work

5.5 House Price Prediction using Machine Learning

The House Price Prediction project is a regression analysis project designed to predict housing prices from property-related features. The project repository contains a housing dataset with 545 rows and 13 columns. Features include area, bedrooms, bathrooms, stories, main-road access, guestroom availability, basement availability, hot-water heating, air conditioning, parking, preferred area, and furnishing status. The target variable is house price.

5.5.1 Dataset and Preparation

The project documentation reports that the dataset contains no missing values. The categorical variables describe property characteristics such as yes/no facilities and furnishing status, while numerical variables include area, bedrooms, bathrooms, stories, and parking.

5.5.2 Regression Model

A linear regression model was trained to predict house prices. The reported model evaluation values were MAE = 1,265,275.67, MSE = 2,750,040,479,309.05, RMSE = 1,658,324.60, and $R^2 = 0.456$.

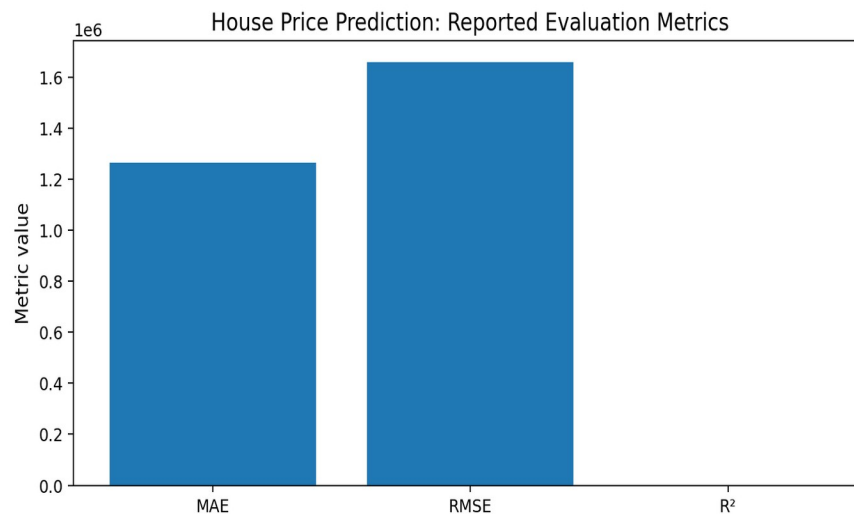


Figure 5.5: Reported evaluation metrics for the House Price Prediction project.

5.5.3 Interpretation

The reported R^2 score of 0.456 indicates that the linear model explains approximately 45.6% of the variance represented by the target variable in the documented evaluation. The project documentation also identifies feature engineering and more advanced models as possible directions for improvement.

5.6 Breast Cancer Diagnosis using Machine Learning

The Breast Cancer Diagnosis project is a supervised classification project using the Breast Cancer Wisconsin (Diagnostic) dataset. The repository documentation describes 569 rows and 31 columns, with 30 computed features and a target representing diagnosis. The target distribution is approximately 37% malignant and 63% benign, and the documentation reports no missing values.

5.6.1 Classification Task

The objective of the project is to classify observations into malignant and benign categories using a machine learning classifier. The model was evaluated using a confusion matrix and a classification report.

5.6.2 Model Performance

The reported confusion matrix is $\begin{bmatrix} 41 & 1 \\ 1 & 71 \end{bmatrix}$. The reported overall accuracy is 0.9825 (98.25%). For the malignant class, precision, recall, and F1-score are each reported as 0.98. For the benign class, precision, recall, and F1-score are each reported as 0.99.

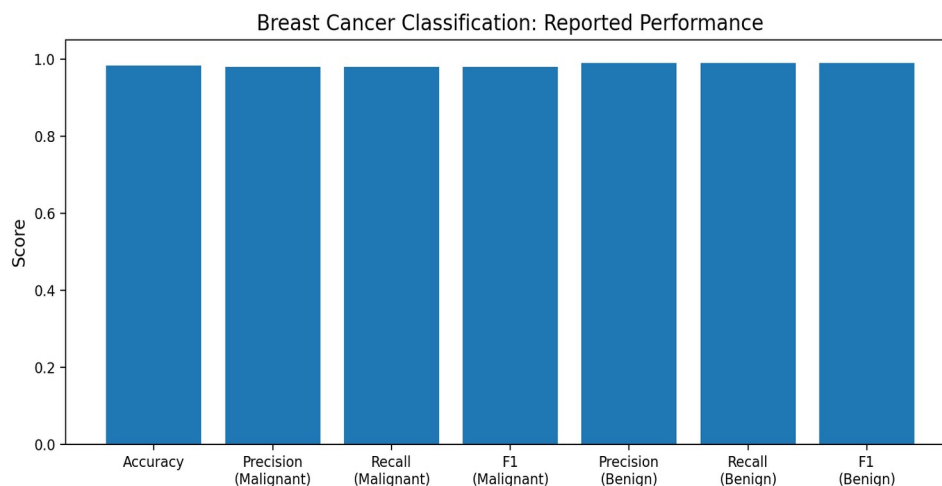


Figure 5.6: Reported performance measures for the Breast Cancer Classification project.

5.6.3 Interpretation

The results indicate strong classification performance in the documented experiment, with balanced precision and recall across both classes. The project documentation notes that feature selection, ensemble methods, or deep learning could be explored as future improvements.

6. MACHINE LEARNING METHODOLOGY

6.1 Problem Definition

The first stage of the machine learning workflow is to convert a real-world question into a predictive problem. In the internship projects, the questions were framed as a continuous-value prediction problem for house prices and a categorical classification problem for breast cancer diagnosis.

6.2 Data Understanding

Data understanding involves examining the available variables, target definition, data types, dataset size, summary statistics, and possible quality issues. This stage helps determine the appropriate preprocessing and modeling strategy.

6.3 Data Preparation

Data preparation includes checking for missing values, understanding numerical and categorical variables, selecting useful features, and preparing the data in a form suitable for model training. The documented datasets used in the two principal ML projects report no missing values.

6.4 Model Training

The House Price Prediction project used linear regression for a continuous target. The Breast Cancer project used a supervised classification approach for malignant-versus-benign prediction.

6.5 Evaluation

Regression performance was evaluated using MAE, MSE, RMSE, and R^2 . Classification performance was evaluated using a confusion matrix, accuracy, precision, recall, and F1-score.

6.6 Interpretation and Improvement

Model evaluation is not the final stage. Results must be interpreted in context and opportunities for improvement identified. The project documentation suggests feature engineering and advanced models for the housing project, and feature selection, ensemble methods, or deep learning for the breast cancer project.

7. RESULTS AND DISCUSSION

Project	Problem Type	Dataset	Main Method	Reported Result
House Price Prediction	Regression	545×13	Linear Regression	$R^2 = 0.456$
Breast Cancer Diagnosis	Classification	569×31	Supervised ML classifier	Accuracy = 98.25%
SQL Student Database	Database	20 sample records	SQL queries	Successful CRUD/query practice
Power BI Dashboard	Visualization	Booking/revenue data	Interactive report	Trend and revenue visualization
Python Calculator	Programming	User input	Conditional logic and loops	Arithmetic operations

7.1 Comparative Discussion

The projects demonstrate a progression from fundamental programming and database operations to analytical visualization and machine learning. The calculator established basic programming logic; SQL introduced structured data storage and retrieval; Power BI developed visual communication skills; and the two machine learning projects introduced predictive modeling and formal evaluation.

The difference between the two principal ML projects also illustrates the distinction between regression and classification. House price prediction produces a continuous numerical output, whereas breast cancer diagnosis produces a categorical prediction. Consequently, the evaluation metrics and interpretation methods differ between the projects.

8. SKILLS AND LEARNING OUTCOMES

8.1 Technical Skills

- Python programming and control-flow logic
- Data manipulation using Pandas
- Numerical computation using NumPy
- Data visualization and graphical interpretation
- SQL database design and querying
- Power BI dashboard development
- Basics and advanced excel learnings
- Supervised machine learning
- Regression and classification
- GitHub-based project documentation

8.2 Professional and Analytical Skills

- Breaking a larger problem into manageable stages
- Interpreting datasets before selecting a model
- Comparing model results using appropriate metrics
- Documenting technical work clearly
- Using visualizations to communicate patterns
- Identifying limitations and possible future improvements

9. CONCLUSION AND FUTURE SCOPE

9.1 Conclusion

The internship at Unlox provided practical exposure to the data science lifecycle, beginning with programming fundamentals and progressing to database operations, visualization, and machine learning. The projects helped convert theoretical concepts into working implementations and improved the ability to work with structured data, predictive models, evaluation metrics, and technical documentation.

The two principal machine learning projects demonstrated practical application of both regression and classification. The House Price Prediction project achieved a documented R^2 score of 0.456 using linear regression, while the Breast Cancer Diagnosis project reported 98.25% accuracy with strong precision, recall, and F1-scores. Together, the projects provided a useful foundation for more advanced machine learning work.

9.2 Future Scope

- Apply feature engineering and feature selection to improve predictive performance.
- Compare multiple regression algorithms for house price prediction.
- Explore ensemble methods for classification.
- Investigate hyperparameter tuning and cross-validation.
- Explore deep learning for appropriate datasets and use cases.
- Deploy trained models through a simple web or API interface.
- Improve dashboard interactivity and connect dashboards to live data sources.
- Build larger end-to-end projects combining database, analytics, visualization, and machine learning.

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